

# AFAGH MEHRI SHERVEDANI (SHE/HER)

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## EDUCATION

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- Ph.D., Electrical and Computer Engineering** 2019 - 2025  
University of Illinois Chicago (UIC), USA.  
Advisor: Professor Miloš Žefran, Thesis: “Reinforcement Learning for Multimodal Human-Robot Interaction”
- Bachelor of Science, Electrical Engineering** 2013 - 2018  
Sharif University of Technology, Tehran, Iran.

## FIELDS OF INTEREST

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Robotics, Artificial Intelligence, Machine Learning, Reinforcement Learning, LLMs/Foundation Models, and Control Systems.

## SKILLS

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**Robotic Platforms.** ROS (Robotic Operating System), AI2-THOR Simulator, Gazebo, Turtlebot3, UR10e Robot (Universal Robots), AlphaBot Robot (Waveshare), NAO Robot (Aldebaran Robotics), Baxter Robot (ReThink Robotics), Raspberry Pi, IMU, RFT Sensor, Microsoft Kinect

**Programming Languages/API.** Python (TensorFlow, Keras, PyTorch, OpenCV, Jupyter NB, Google Colab), Google Cloud API, OpenAI API, Docker, GitHub, MATLAB/Simulink, C/C++, SQL

## HONORS AND AWARDS

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- UIC Electrical and Computer Engineering Graduate Student Award, 3000\$ grant awarded to top admitted graduate students. September 2019
- Ranked in the top 1% nationwide, in the Iranian Universities National Entrance Exam, Iran. June 2013

## RESEARCH EXPERIENCE

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**Research Assistant**, Robotics Lab, ECE Department, UIC, Chicago, IL Aug. 2019 - Spring 2025  
Supervisor: Professor Miloš Žefran

- Developed and enhanced a data-driven **Neural Network-based simulator** acting as a human in a **goal-oriented multimodal task**. (Publication [C3](#).)
- Developed an AI agent, based on **deep Q-learning algorithm**, as an assistive robot in a **goal-oriented multimodal task**. (Publication [J1](#).)
- Developed an **ML-based** human intent recognition module, optimized the **object localization/pose estimation** module, and implemented the **kinematics measurement** module in a physical human-robot collaboration task. (Publication [C1](#).)

**Intern**, Robot Intelligence through Perception Lab, Toyota Technological Institute at Chicago (TTIC), Chicago, IL Summer 2024  
Host: Professor Matthew R. Walter

- Developed an **LLM-based hybrid planning framework** that integrates LLMs with iterative **human feedback** for human-robot collaboration. (Publication [U1](#).)

**Intern**, Alphanumeric Lab, Department of Biomedical Informatics, Emory University, Atlanta, GA Summer 2021  
Host: Professor Reza Sameni

- Developed signal processing algorithms in Python to extract ECG signal data points from colored ECG images. (Publication [J2](#).)

**Intern**, Maharban Company, Tehran, Iran Summer 2016

- Developed an algorithm based on graph theory to minimize the required time for automatic cyclic check of logical control sequences.

## PUBLICATIONS

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### Journal

- J1. **A. Mehri Shervedani**, S. Li, N. Monaikul, B. Abbasi, M. Žefran, B. Di Eugenio, “**Multimodal Reinforcement Learning for Robots Collaborating with Humans**”, International Journal of Social Robotics (2025), 1–23.

- J2. K. Kodthalu Shivashankara, **A. Mehri Shervedani**, G. D. Clifford, M. A. Reyna, R. Sameni, “**ECG-Image-Kit: A Synthetic Image Generation Toolbox to Facilitate Deep Learning-Based Electrocardiogram Digitization**”, Physiological Measurement, 45, 055019 (2024).

#### Conference

- C1. Z. Rysbek, S. Li, **A. Mehri Shervedani**, M. Žefran “**Proactive Robot Control for Collaborative Manipulation Using Human Intent**”, IEEE ICRA 2024, Yokohama, Japan, May 2024.
- C2. S. Li, **A. Mehri Shervedani**, M. Žefran, I. Paprotny, “**Group-Control Motion Planning Framework for Microrobot Swarms in a Global Field**”, The 16th Workshop on the Algorithmic Foundations of Robotics (WAFR), Chicago, USA, Oct 2024.
- C3. **A. Mehri Shervedani**, S. Li, N. Monaikul, B. Abbasi, B. Di Eugenio, M. Žefran, “**An End-to-End Human Simulator for Task-Oriented Multimodal Human-Robot Collaboration**”, IEEE RO-MAN 2023, Busan, South Korea, Aug 2023.

#### Workshop

- W1. **A. Mehri Shervedani** et al., “**Evaluating Multimodal Interaction of Robots Assisting Older Adults**”, ICSR 2022 Workshop on Evaluating Social Assistive Robots in Healthcare, 14th International Conference on Social Robotics, Florence, Italy, 2022.

#### Under Review

- U1. **A. Mehri Shervedani**, K. Oh, S. Li, et al. “**High-Level Instructions to Task Plans: A Feedback-Driven Human-Robot Collaboration Task Planning with LLMs**”, IEEE ICRA 2026, Vienna, Austria, June 2026.

### TEACHING EXPERIENCE

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**Instructor**, ECE Department, UIC, Chicago, IL

Spring 2025

ECE 452 - Robotics: Algorithms/Control

**Teaching Assistant**, ECE Department, UIC, Chicago, IL

Aug. 2019 - Dec. 2024

ECE 417 - Digital Signal Processing II | ECE 350 - Principles of Automatic Control | ECE 333 - Computer Communication Networks I

### PROFESSIONAL DEVELOPMENT IN TEACHING

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Completed six **OneHE** courses on evidence-based teaching and student support: Improving the Student Experience with Exit Tickets; The Role of Structure in Inclusive Teaching; Small Teaching: Learning Through Connecting; Supporting Students in Developing Information Literacy; Applying the Science of Motivation to Teaching; Leading In-Class Group Work.

### MENTORING EXPERIENCE

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**Intern Supervisor**, ECE Department, UIC, Chicago, IL

July. 2023 - Aug 2023

- Defined and supervised two undergraduate robotics projects: one on human-pointing gesture recognition with an RGB-D Kinect camera, and another on Baxter robot motion planning and inverse kinematics. Mentored students in technical skills, problem-solving, and teamwork.

**Undergrad Research Project Supervisor**, ECE Department, UIC, Chicago, IL

Jan. 2022 - May 2022

- Supervised four undergraduates on an LSTM-based neural network project for classifying haptic actions from video snippets. Taught principals of neural networks, LSTMs, object-oriented programming, and interactive Python programming to the team.

### SERVICE

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#### Reviewer

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025
- IEEE International Conference on Robotics and Automation (ICRA) 2024
- IEEE Access Journal, 2023

#### Organizer

- Robotics Lab Outreach Presenter**, ECE Department, UIC, Chicago, IL Sept 2022, Sept 2023  
Presented lab research and live robotics demonstrations to visiting high-school students as part of department recruitment events.
- Vice Chair**, Water and Energy Nexus Conference, Sharif University of Technology, Iran. Jun 2015 - Mar 2016
- Cultural Vice Chair**, Water Event, Sharif University of Technology and University of Tehran, Iran. Oct 2015
- Active member of “Sharif Green Society”, Sharif University of Technology, Tehran, Iran. 2014 - 2018